

Week 1, Lecture 3 Handout: Interactive Elements, Media, & Form Architecture

Layout Containers, Media, & Interactive Forms (The Final 11 Tags)

1. Document Layout & Styling Containers

These tags are essential because they dictate how layout boxes and assets are targeted, organized, and arranged using CSS next week.

- **<nav>**
Explicitly defines a structural block intended to hold major navigation menus or link collections. Search engines and screen readers rely heavily on this tag to understand how to move around a website. [1, 2, 3, 4, 5]
- ****
Embeds a graphic image into the webpage. It is a self-closing element that requires a source (`src`) path to load the file and alternate text (`alt`) to describe the image for web accessibility and SEO. [6]
- **<div>**
A generic, invisible block-level container used to group elements together. It carries zero structural meaning, making it the single most important tag for structural CSS layouts like Flexbox and Grid. [7]
- ****
A generic inline container used to isolate specific words or phrases inside text blocks (like inside a `<p>` tag) for targeted CSS styling.

2. Form Architecture & Data Intakes

Modern frontend applications revolve around user data. These tags form the complete architecture required to build clean, functional user input fields.

- **<form>**
The master container that wraps all interactive fields. It captures user inputs and prepares the compiled data to be processed or sent to a database.
- **<label>**
Provides a clear text description for a specific input field. Clicking a label automatically shifts the user's cursor focus directly inside its linked input box, which is vital for web accessibility.
- **<input>**
A highly versatile, self-closing data field. Its input style changes completely based on its `type` attribute (e.g., `type="text"` for names, `type="email"` for electronic addresses, or `type="password"` to mask typing).
- **<textarea>**
Creates an expandable, multi-line text input field designed to let users write long-form entries like feedback, reviews, or messages.
- **<select>**
Defines a clickable dropdown menu, allowing users to safely choose an answer from a pre-defined list of choices.

- **<option>**
Represents the individual, selectable choices nested tightly inside a **<select>** dropdown menu container.
- **<button>**
A clickable user interface element used to submit compiled form data to a server or fire active browser behaviors. [8, 9, 10, 11, 12]

3. Numbered Sequencing

- ** (Ordered List)**
Defines a sequential list of items. Unlike the bulleted unordered list (****), the browser automatically numbers each child list item (****) with ascending numbers (1, 2, 3...) by default.

HTML Capstone Project: Comprehensive Profile & Application Page

Objective: Combine every single tag learned throughout Week 1 (Lectures 1, 2, and 3) into a single, comprehensive, unstyled web layout to officially complete your HTML foundations.

Task Instructions

1. Open **VS Code** and create a brand new file named **project.html**.
2. Initialize your page with the valid HTML5 core skeleton (**<!DOCTYPE html>**, **<html>**, **<head>**, **<title>**, **<body>**) and change the title to your project name.
3. **The Header & Navigation Zone:**
 - Wrap your initial heading block in a **<header>** tag.
 - Add a site title using an **<h1>** tag.
 - **Open a **<nav>** tag directly below the heading.** Inside, build an unordered list (****) with three list items (****) containing links (**<a>**) for *Home*, *About*, and *Contact*. [13]
4. **The Main Content Area:**
 - Wrap your primary content body inside a **<main>** container.
 - Create a **<section>** explaining your application values. Inside, write a short paragraph (**<p>**) with one word wrapped in **** and another in ****.
 - Embed a professional image using the **** tag with a valid **src** and **alt** description.
 - Create an ordered list (****) detailing 3 numbered steps to submit an application. [14]
5. **The Interactive Form Zone:**
 - Create a second **<section>** called "Application Form".
 - Inside, open a **<form>** element.
 - Use **<div>** containers to keep your fields neatly organized on fresh, separate lines.

- Add text inputs for Name and Email using `<input>`, complete with their matching `<label>` elements.
- Add a `<select>` dropdown with three `<option>` elements for selecting an industry role.
- Add a `<textarea>` for a brief message or cover letter.
- Add a final form submission `<button>` containing the text "Submit Application".

6. The Footer Zone:

- At the very base of your document, add a `<footer>` enclosing a copyright line and a working anchor link (`<a>`) that connects back to your personal portfolio page or GitHub account. [15]

7. Save your document and launch it via **Live Server** to ensure every link, form input, and dropdown option operates correctly before starting CSS next week!
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